Chapter 23

Machine Learning in Mass Spectrometry: High-Dimensional Regularization, Tree Ensembles, and Resampling Integrity

Lucas VHH TRAN[1]

Abstract

While classical differential abundance analysis addresses univariate hypothesis testing across individual proteins or metabolites, clinical and translational mass spectrometry increasingly demands multivariate predictive modeling. The primary translational objective is to construct generalizable molecular signatures capable of patient stratification, early disease diagnosis, and therapeutic response forecasting. However, quantitative mass spectrometry resides squarely within the ultra-high-dimensional regime where the number of quantified molecular features exceeds the sample cohort size by several orders of magnitude ($p \gg n$; frequently $p \sim 10^3$–10^5 versus $n \sim 10^1$–10^2). In this mathematical territory, standard statistical estimators collapse, ordinary least squares is singular, and empirical classification accuracy can trivially reach 100% on random noise through geometric over-fitting.

This chapter establishes the theoretical foundations and computational architecture required to build, tune, and validate rigorous machine learning models in mass spectrometry. We examine the geometric pathologies of high-dimensional Euclidean space, the Curse of Dimensionality, and Cover's theorem on the linear separability of random patterns. We derive the analytical formulations of penalized loss functions, with special focus on the Elastic Net and its grouping property for co-eluting, highly correlated peptides and metabolites. We dissect non-linear ensemble algorithms, including Random Forests and Gradient Boosted Decision Trees (XGBoost), analyzing their bias-variance trade-offs, and compare Gini impurity against game-theoretic SHAP feature importance. Crucially, we formalize the topology of data leakage and formulate nested cross-validation, class-imbalance corrections, probability calibration (Brier score and reliability curves), and clinical decision curve analysis. The chapter culminates in an end-to-end reproducible R laboratory using the modern `tidymodels` *ecosystem, the "Five Deadly Sins" of omics machine learning, a tutor diagnostic heuristics checklist, and comprehensive multi-tier exercises with complete worked solutions.*

Keywords: Machine Learning, Elastic Net, High-Dimensional Regularization, Random Forest, XGBoost, SHAP Values, Data Leakage, Nested Cross-Validation, Probability Calibration, Tidymodels.

[1]CK-CARE Davos, Christine Kühne – Center for Allergy Research and Education, Davos, Switzerland. Email: `lucas.tran@ck-care.ch`

23.1 Tutor's Methodological Roadmap for Omics Machine Learning

Before writing machine learning code or executing hyperparameter grids, the translational mass spectrometrist must understand the multi-stage machine learning workflow. Table 23.1 summarizes the four essential checkpoints required to build publication-grade, generalizable predictive signatures.

Table 23.1: Tutor's Methodological Roadmap: Four Checkpoints for Machine Learning in Mass Spectrometry

Stage	Physical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Data Topology	High-dimensional space ($p \gg n$) causes distance concentration and empty space [10, 17].	Audit feature missingness, perform group-preserving train/test splits.	Pre-screening features on whole cohort (data leakage). Remedy: Perform all filtering inside CV folds [1].
2. Model Architecture	Linear sparsity (Elastic Net [30]) vs. non-linear interactions (Random Forest, XGBoost [4, 9]).	Select inductive bias matching biological hypothesis; preserve collinear groups.	LASSO randomly dropping correlated peptides. Remedy: Elastic Net ($\alpha \in [0.2, 0.8]$) [30].
3. Resampling Integrity	Hyperparameter tuning on test splits produces optimistic performance bias [19].	Implement nested cross-validation (inner tuning loop, outer evaluation loop).	Over-optimistic AUC reports. Remedy: Nested 5×5 cross-validation with outer test holdout [20].
4. Clinical Utility	Raw classifier scores are uncalibrated risk probabilities [5, 23].	Evaluate Brier score, Expected Calibration Error (ECE), and Decision Curve Analysis.	High AUC with severely distorted probabilities. Remedy: Platt scaling or isotonic regression [29].

Methodological Note

Tutor's Methodological Warning: The Data Leakage Catastrophe
Data leakage is the single most common cause of retracted machine learning omics papers [1, 6]. If you normalize, impute, or screen features using all N patients before cross-validation splitting, information from the validation test fold leaks into the training pipeline. The resulting 98% accuracy is an artifact of leakage that collapses to 50% on independent clinical cohorts. Always embed all preprocessing inside the cross-validation recipe [20].

23.2 The High-Dimensional Pathology ($p \gg n$) in Mass Spectrometry

Translational mass spectrometry datasets are characterized by extreme dimensional asymmetry. A typical clinical proteomics experiment quantifies $p = 5{,}000$ to $12{,}000$ proteins, and deep metabolomics profiles thousands of spectral features across $n = 50$ to 200 patients. This condition violates the foundational asymptotic regularity assumptions of classical multivariate statistics.

23.2.1 The Curse of Dimensionality and Geometric Sparsity

As dimension p increases, the geometry of Euclidean space undergoes counter-intuitive distortions:

1. **Hypersphere Volume Vanishing:** The volume of a p-dimensional hypersphere of radius r inscribed within a hypercube of side length $2r$ is:

$$V_p(r) = \frac{\pi^{p/2}}{\Gamma\left(\frac{p}{2}+1\right)} r^p \tag{23.1}$$

   The ratio of the volume of the sphere to the volume of the enclosing hypercube $(2r)^p$ converges asymptotically to zero:

$$\lim_{p\to\infty} \frac{V_p(r)}{(2r)^p} = \lim_{p\to\infty} \frac{\pi^{p/2}}{2^p \Gamma\left(\frac{p}{2}+1\right)} = 0 \tag{23.2}$$

   Almost all volume in high-dimensional space concentrates in the "corners" of the hypercube [17].

2. **Distance Concentration:** For n points drawn independently from an arbitrary probability distribution in $\mathbb{R}^p$, the distance between any two points concentrates around the mean distance:

$$\lim_{p\to\infty} \frac{\text{dist}_{\max} - \text{dist}_{\min}}{\text{dist}_{\min}} = 0 \tag{23.3}$$

   In ultra-high dimensions, all sample pairs become virtually equidistant from one another, causing unpenalized nearest-neighbor algorithms to degenerate.

23.2.2 Cover's Theorem on Separability

Consider n random data points in general position in $\mathbb{R}^p$, each assigned a binary label $y_i \in \{-1, +1\}$ uniformly at random. By Cover's theorem [10], the probability $P(n,p)$ that these points are linearly separable by a hypercube or hyperplane is:

$$P(n,p) = \left(\frac{1}{2}\right)^{n-1} \sum_{k=0}^{p} \binom{n-1}{k} \tag{23.4}$$

Whenever $p \geq n-1$, the summation encompasses all terms $0 \leq k \leq n-1$, yielding:

$$\sum_{k=0}^{n-1} \binom{n-1}{k} = 2^{n-1} \implies P(n,p) = 1.0 \tag{23.5}$$

Biological Discovery & Validation

The Mathematical Inevitability of 100% Training Accuracy:
When features outnumber samples ($p > n$), an unregularized linear classifier will achieve 100% training accuracy on pure Gaussian noise with randomly assigned patient labels. High training accuracy is never evidence of biological biomarker discovery; it is a mathematical property of the linear system's null space [10, 28].

23.3 Penalized Estimation: Ridge, LASSO, and the Elastic Net

To prevent over-fitting in the $p \gg n$ regime, we introduce structural regularization by augmenting the empirical negative log-likelihood with a penalty functional $\mathscr{P}(\beta)$ on the coefficient vector [18, 27].

For a binary classification problem with patient labels $y_i \in \{0,1\}$ and standardized feature vector $\mathbf{x}_i \in \mathbb{R}^p$, the penalized logistic regression objective is:

$$\min_{\beta_0,\beta} \left\{ -\frac{1}{n}\sum_{i=1}^{n} [y_i \ln \pi(\mathbf{x}_i) + (1-y_i)\ln(1-\pi(\mathbf{x}_i))] + \lambda\, \mathscr{P}_\alpha(\beta) \right\} \tag{23.6}$$

where $\pi(\mathbf{x}_i) = \frac{1}{1+\exp(-(\beta_0+\mathbf{x}_i^T\beta))}$.

23.3.1 The Elastic Net Penalty

Zou and Hastie [30] formulated the Elastic Net penalty as a convex combination of the ℓ_1 (LASSO [27]) and ℓ_2 (Ridge [18]) penalties:

$$\mathscr{P}_\alpha(\beta) = \alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2 = \sum_{j=1}^{p}\left(\alpha|\beta_j| + \frac{1-\alpha}{2}\beta_j^2\right) \tag{23.7}$$

where $\lambda > 0$ dictates the global shrinkage intensity, and $\alpha \in [0,1]$ governs the penalty geometry:

- $\alpha = 0$ corresponds to **Ridge Regression** (ℓ_2), shrinking coefficients toward zero without performing feature selection ($\beta_j \neq 0\ \forall j$).
- $\alpha = 1$ corresponds to the **LASSO** (ℓ_1), inducing exact mathematical sparsity by setting irrelevant features to zero.
- $0 < \alpha < 1$ represents the **Elastic Net**, combining sparsity with the grouping of collinear features.

23.3.2 Coordinate Descent and the Soft-Thresholding Operator

Optimization of Eq. (23.6) is efficiently solved via cyclical coordinate descent [14]. At each step, fixing all parameters β_k ($k \neq j$), the partial derivative with respect to β_j updates via the bivariate soft-thresholding operator:

$$\hat{\beta}_j = \frac{\mathscr{S}\left(\frac{1}{n}\sum_{i=1}^{n} x_{ij}(y_i - \tilde{y}_i^{(-j)}),\ \lambda\alpha\right)}{1+\lambda(1-\alpha)} \tag{23.8}$$

where the soft-thresholding function $\mathscr{S}(z,\gamma)$ is defined as:

$$\mathscr{S}(z,\gamma) = \text{sign}(z)\max(0, |z| - \gamma) \tag{23.9}$$

Whenever the partial correlation $|z| \leq \lambda\alpha$, the feature coefficient is set identically to zero.

23.3.3 Analytical Proof of the Grouping Property

In mass spectrometry, molecular features often exhibit strong collinearity (e.g., peptide fragments from the same protein, charge states of identical peptides, or metabolites in the same metabolic pathway). The LASSO possesses a well-known pathology: given a cluster of highly correlated features ($r \approx 1$), it randomly selects one feature and sets all others to zero.

The Elastic Net mathematically enforces the *grouping property*: correlated features enter or leave the model together.

Mathematical Deep Dive

Mathematical Proof of the Elastic Net Grouping Bound:
Let $\hat{\beta}$ be the Elastic Net minimizer for normalized predictors $\mathbf{x}_j$ and $\mathbf{x}_k$ with sample correlation $\rho = \mathbf{x}_j^T \mathbf{x}_k$. Assume $\text{sign}(\hat{\beta}_j) = \text{sign}(\hat{\beta}_k) > 0$. By the sub-gradient optimality conditions of Eq. (23.6):

$$-\frac{1}{n}\mathbf{x}_j^T(\mathbf{y} - \mathbf{X}\hat{\beta}) + \lambda\alpha + \lambda(1-\alpha)\hat{\beta}_j = 0 \tag{23.10}$$

$$-\frac{1}{n}\mathbf{x}_k^T(\mathbf{y} - \mathbf{X}\hat{\beta}) + \lambda\alpha + \lambda(1-\alpha)\hat{\beta}_k = 0 \tag{23.11}$$

Subtracting the second equality from the first:

$$\lambda(1-\alpha)(\hat{\beta}_j - \hat{\beta}_k) = \frac{1}{n}(\mathbf{x}_j - \mathbf{x}_k)^T(\mathbf{y} - \mathbf{X}\hat{\beta}) \tag{23.12}$$

Taking the absolute value and applying the Cauchy-Schwarz inequality:

$$|\hat{\beta}_j - \hat{\beta}_k| \leq \frac{1}{\lambda(1-\alpha)} \frac{1}{\sqrt{n}} \|\mathbf{x}_j - \mathbf{x}_k\|_2 \cdot \frac{1}{\sqrt{n}} \|\mathbf{y} - \mathbf{X}\hat{\beta}\|_2 \tag{23.13}$$

Since $\mathbf{x}_j$ and $\mathbf{x}_k$ are standardized ($\|\mathbf{x}_j\|_2^2 = \|\mathbf{x}_k\|_2^2 = n$), we have $\|\mathbf{x}_j - \mathbf{x}_k\|_2 = \sqrt{2n(1-\rho)}$. Thus:

$$|\hat{\beta}_j - \hat{\beta}_k| \leq \frac{\sqrt{2(1-\rho)}}{\lambda(1-\alpha)} \cdot \text{RMSE} \tag{23.14}$$

As correlation $\rho \to 1$, the upper bound $|\hat{\beta}_j - \hat{\beta}_k| \to 0$. Correlated biomarkers receive near-identical coefficients, preventing arbitrary feature selection [30].

23.3.4 Tree Ensembles and Interpretability: Random Forest, XGBoost, and SHAP

When biological relationships are non-linear (e.g., threshold effects or pathway epistatic interactions), tree ensemble methods excel:

- **Random Forests:** Breiman [4] constructs hundreds of de-correlated decision trees using bootstrap aggregation (bagging) and random feature sub-sampling ($\sqrt{p}$ features per split). While robust to noise, classical Mean Decrease Gini importance suffers from severe selection bias toward continuous, high-variance noise features [26].
- **Gradient Tree Boosting (XGBoost):** Friedman [13] and Chen & Guestrin [9] fit trees sequentially to the negative gradient of the loss function with second-order Taylor expansion and leaf-weight shrinkage.
- **SHAP (SHapley Additive exPlanations):** Grounded in cooperative game theory, Lundberg et al. [21, 22] formulated TreeSHAP to calculate the exact marginal contribution of each protein to individual patient predictions, eliminating the heuristic biases of Gini impurity.

23.3.5 Model Probability Calibration and Decision Curve Analysis

In clinical medicine, classification labels (0 or 1) are insufficient; physicians make treatment decisions based on predicted disease probabilities $\hat{p}_i \in [0,1]$.

- **Brier Score and ECE:** The Brier score [5] measures probability error: $\text{BS} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{p}_i)^2$. The Expected Calibration Error (ECE) computes the weighted gap between predicted confidence and true positive rate across probability bins [23].

- **Decision Curve Analysis (DCA):** Vickers & Elkin [29] introduced net benefit analysis:

$$\text{Net Benefit} = \frac{\text{True Positives}}{N} - \frac{\text{False Positives}}{N}\left(\frac{p_t}{1-p_t}\right) \tag{23.15}$$

  where p_t represents the clinical decision threshold probability. DCA directly benchmarks the clinical net utility of the mass spectrometry signature against treat-all and treat-none strategies.

23.4 Hands-On R Laboratory: Tidymodels Workflow in Proteomics

We now implement an end-to-end reproducible machine learning pipeline using the modern `tidymodels` architecture [20, 19].

R Lab Listing 1: Simulating High-Dimensional Biomarker Discovery Cohort [R 4.6 / Bioconductor]

```
suppressPackageStartupMessages({
  library(tidymodels)
  library(glmnet)
  library(dplyr)
  library(ggplot2)
})

set.seed(123)
n_patients <- 80
n_proteins <- 600

# True biological label: 40 Control, 40 Tumor
status <- factor(rep(c("Control", "Tumor"), each = 40), levels = c("Control", "Tumor"))

# Simulate baseline expression
expr_data <- matrix(rnorm(n_patients * n_proteins, mean = 20, sd = 1.5),
                    nrow = n_patients, ncol = n_proteins)
colnames(expr_data) <- paste0("PROT_", sprintf("%03d", 1:n_proteins))

# Inject true signature: 10 collinear proteins upregulated in Tumor (+2.2 log2FC)
de_genes <- 1:10
expr_data[41:80, de_genes] <- expr_data[41:80, de_genes] + 2.2

df_all <- as.data.frame(expr_data)
df_all$Status <- status

cat(sprintf("Simulated cohort: %d patients x %d proteins.\n", n_patients, n_proteins))
cat(sprintf("True biomarker signature: Proteins 1 to 10 (collinear block).\n"))
```

```
Simulated cohort: 80 patients x 600 proteins. True biomarker signature: Proteins 1 to 10 (collinear block).
```

R Lab Listing 2: Strict Cross-Validation Pipeline with Recipe Architecture [R 4.6 / Bioconductor]

```r
# 1. Stratified 75/25 Train-Test Split
set.seed(42)
data_split <- initial_split(df_all, prop = 0.75, strata = Status)
train_data <- training(data_split)
test_data  <- testing(data_split)

# 2. Preprocessing Recipe: Normalization strictly inside training folds
rec <- recipe(Status ~ ., data = train_data) %>%
  step_normalize(all_predictors())

# 3. Model Specification: Elastic Net (alpha = 0.5)
enet_spec <- logistic_reg(penalty = tune(), mixture = 0.5) %>%
  set_engine("glmnet") %>%
  set_mode("classification")

# 4. Workflow Container
enet_wf <- workflow() %>%
  add_recipe(rec) %>%
  add_model(enet_spec)

# 5. 5-Fold Cross-Validation for Hyperparameter Tuning
set.seed(42)
folds <- vfold_cv(train_data, v = 5, strata = Status)
param_grid <- grid_regular(penalty(range = c(-3, 0)), levels = 10)

tune_res <- tune_grid(
  enet_wf,
  resamples = folds,
  grid = param_grid,
  metrics = metric_set(roc_auc, accuracy)
)

# Extract best model and evaluate on untouched holdout test set
best_param <- select_best(tune_res, metric = "roc_auc")
final_wf   <- finalize_workflow(enet_wf, best_param)
final_fit  <- last_fit(final_wf, data_split)

test_metrics <- collect_metrics(final_fit)
cat(sprintf("Optimal Elastic Net Penalty (lambda): %.4f\n", best_param$penalty))
cat(sprintf("Independent Holdout Test AUC:         %.3f\n",
            test_metrics$mean[test_metrics$.metric == "roc_auc"]))
cat(sprintf("Independent Holdout Test Accuracy:    %.3f\n",
            test_metrics$mean[test_metrics$.metric == "accuracy"]))
```

```
Optimal Elastic Net Penalty (lambda): 0.1000 Independent Holdout Test AUC: 0.960 Independent Holdout Test Accuracy: 0.950
```

23.5 The Five Deadly Sins of Omics Machine Learning

Methodological Pitfall & Deadly Sin

Deadly Sin 1: The Pre-Filtering Data Leakage Catastrophe.
Mechanism: Screening features across all N patients using univariate t-tests ($p < 0.05$) before splitting into training and validation sets.
Consequence: The validation set information leaks into the feature selector, producing 99% cross-validation accuracy on pure random noise that completely fails in clinical validation [1].
Remedy: Execute all feature filtering, scaling, and imputation strictly inside the cross-validation training fold (`tidymodels::recipe`) [20].

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Using Default LASSO ($\alpha = 1$) on Strongly Collinear Omics Data.
Mechanism: Applying standard LASSO to co-eluting peptides or tightly co-regulated metabolites.
Consequence: LASSO arbitrarily picks one feature and zeroes out all biological partners, leading to erratic signatures that vary wildly between cross-validation folds [30].
Remedy: Deploy the Elastic Net with $\alpha \in [0.2, 0.8]$ to enforce the grouping property (Eq. 23.14).

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Applying SMOTE to the Entire Dataset Before Resampling.
Mechanism: Generating synthetic minority samples using SMOTE on the complete dataset before cross-validation.
Consequence: Synthetic test points are derived directly from training neighbors, guaranteeing massive over-fitting.
Remedy: Apply SMOTE strictly to the training fold within the resampling recipe [8].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Reporting Unnested Cross-Validation Performance.
Mechanism: Tuning hyperparameters on a cross-validation grid and reporting the best-tuned performance as the model's true accuracy.
Consequence: Hyperparameter tuning on validation folds introduces optimistic selection bias.
Remedy: Deploy nested cross-validation (5×5) or an independent untouched holdout test set [19].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Trusting Uncalibrated Machine Learning Probabilities.
Mechanism: Interpreting raw Random Forest vote fractions or SVM margins as absolute clinical risk probabilities.
Consequence: Random Forest probabilities cluster away from 0 and 1, distorting clinical decision thresholds.
Remedy: Audit Brier scores, plot reliability diagrams, and apply Platt scaling or isotonic calibration [5, 23].

23.6 Tutor's Self-Assessment Diagnostic Checklist

To confirm the methodological validity of your machine learning workflow prior to reporting clinical biomarker signatures, execute the diagnostic audit summarized in Table 23.2.

Table 23.2: Tutor's Self-Assessment Diagnostic Checklist for Omics Machine Learning

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Leakage Audit	Zero preprocessing performed prior to train/test split.	Encapsulate all transformations inside `recipe` pipeline [20].
2	Grouping Bound Check	Collinear biomarker blocks retained collectively.	Shift from LASSO ($\alpha = 1$) to Elastic Net ($\alpha = 0.5$) [30].
3	Resampling Architecture	Nested cross-validation or untouched holdout set ($N \geq 20$).	Eliminate unnested tuning bias; report outer test AUC [19].
4	Probability Calibration	Expected Calibration Error ECE ≤ 0.15; Brier score < 0.20.	Apply Platt scaling or isotonic calibration [23].
5	Clinical Net Benefit	Decision Curve Analysis demonstrates positive net benefit.	Benchmark against treat-all / treat-none strategies [29].

23.7 Chapter Exercises

Exercise 23.1 (Elastic Net Grouping Bound Calculation):
Two co-eluting biomarker peptides have a sample correlation of $\rho = 0.98$. An Elastic Net model is fitted with penalty parameter $\lambda = 0.50$ and mixture $\alpha = 0.40$. The root-mean-squared error of the fit is RMSE $= 0.80$.

1. Compute the theoretical upper bound on the coefficient difference $|\hat{\beta}_1 - \hat{\beta}_2|$ using Eq. (23.14).
2. If the correlation drops to $\rho = 0.50$, what is the new upper bound? Explain the biological implication of this difference.

Exercise 23.2 (Probability Calibration and Brier Score Audit):
A machine learning classifier predicts disease probabilities for $n = 5$ patients:
$\hat{\mathbf{p}} = (0.10, 0.30, 0.60, 0.85, 0.90)^T$. The true biological outcomes are $\mathbf{y} = (0, 0, 1, 1, 1)^T$.

1. Compute the empirical Brier score BS $= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{p}_i)^2$.
2. Compare this to a naive uninformative classifier that predicts $\hat{p} = 0.50$ for all patients.

23.8 Comprehensive Worked-Out Solutions

Solution to Exercise 23.1: Elastic Net Grouping Bound

1. Upper Bound for $\rho = 0.98$:
From Eq. (23.14):

$$|\hat{\beta}_1 - \hat{\beta}_2| \leq \frac{\sqrt{2(1-\rho)}}{\lambda(1-\alpha)} \cdot \text{RMSE}$$

Given $\rho = 0.98$, $\lambda = 0.50$, $\alpha = 0.40$, RMSE $= 0.80$:

$$\sqrt{2(1-0.98)} = \sqrt{2(0.02)} = \sqrt{0.04} = 0.20$$

$$\lambda(1-\alpha) = 0.50(1-0.40) = 0.50(0.60) = 0.30$$

$$|\hat{\beta}_1 - \hat{\beta}_2| \leq \frac{0.20}{0.30} \times 0.80 = \frac{2}{3} \times 0.80 \approx 0.533$$

2. Upper Bound for $\rho = 0.50$:

$$\sqrt{2(1-0.50)} = \sqrt{2(0.50)} = \sqrt{1.0} = 1.00$$

$$|\hat{\beta}_1 - \hat{\beta}_2| \leq \frac{1.00}{0.30} \times 0.80 = 3.333 \times 0.80 \approx 2.667$$

Biological Interpretation: For highly correlated biomarkers ($\rho = 0.98$), the Elastic Net guarantees that their coefficients cannot differ by more than 0.533. When correlation drops to $\rho = 0.50$, the bound relaxes to 2.667 (5× wider), allowing the model to decouple features with independent biological trajectories [30].

Solution to Exercise 23.2: Brier Score Computation

1. Brier Score of Classifier:
Individual squared errors:

$$(y_1 - \hat{p}_1)^2 = (0-0.10)^2 = 0.0100$$
$$(y_2 - \hat{p}_2)^2 = (0-0.30)^2 = 0.0900$$
$$(y_3 - \hat{p}_3)^2 = (1-0.60)^2 = 0.1600$$
$$(y_4 - \hat{p}_4)^2 = (1-0.85)^2 = 0.0225$$
$$(y_5 - \hat{p}_5)^2 = (1-0.90)^2 = 0.0100$$

Sum of squared errors:

$$\sum (y_i - \hat{p}_i)^2 = 0.0100 + 0.0900 + 0.1600 + 0.0225 + 0.0100 = 0.2925$$

$$\text{BS} = \frac{0.2925}{5} = 0.0585$$

2. Comparison to Clueless Benchmark:
For $\hat{p} \equiv 0.50$:

$$(0-0.5)^2 \times 2 + (1-0.5)^2 \times 3 = 0.25 \times 5 = 1.25 \implies \text{BS}_{\text{naive}} = \frac{1.25}{5} = 0.2500$$

The trained classifier achieves a Brier score of 0.0585, an exceptional 76.6% reduction in prediction error compared to the uninformative baseline (0.2500), demonstrating strong clinical probability calibration [5].

Canonical References

Bibliography

[1] Ambroise, C., & McLachlan, G. J. (2002). Selection bias in gene extraction on the basis of microarray gene-expression data. *Proceedings of the National Academy of Sciences*, 99(10), 6562–6566. doi:10.1073/pnas.102102699.

[2] Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.

[3] Boland, M. R. (2023). *Health Data Analytics with R: A Hands-On Guide*. Springer Nature, Cham, Switzerland. doi:10.1007/978-3-031-18099-6.

[4] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. doi:10.1023/A:1010933404324.

[5] Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

[6] Broadhurst, D. I., & Kell, D. B. (2006). Statistical strategies for avoiding false discoveries in metabolomics and related systems biology. *Metabolomics*, 2(4), 171–196. doi:10.1007/s11306-006-0037-z.

[7] Broadhurst, D., Goodacre, R., Reinke, S. N., Kuligowski, J., Wilson, I. D., Lewis, M. R., & Dunn, W. B. (2018). Guidelines and considerations for the use of system suitability and quality control samples in mass spectrometry assays applied in untargeted metabolomic studies. *Metabolomics*, 14(6), 72. doi:10.1007/s11306-018-1367-3.

[8] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357. doi:10.1613/jair.953.

[9] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). doi:10.1145/2939672.2939785.

[10] Cover, T. M. (1965). Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers*, EC-14(3), 326–334. doi:10.1109/PGEC.1965.264137.

[11] Cox, J., Hein, M. Y., Luber, C. A., Parida, I., Cox, J., & Mann, M. (2014). Accurate proteome-wide label-free quantification by delayed normalization and maximal peptide ratio extraction, termed MaxLFQ. *Molecular & Cellular Proteomics*, 13(9), 2513–2526. doi:10.1074/mcp.M113.031591.

[12] Demichev, V., Messner, C. B., Vernardis, S. I., Lilley, K. S., & Ralser, M. (2020). DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput. *Nature Methods*, 17(1), 41–44. doi:10.1038/s41592-019-0638-x.

[13] Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232. doi:10.1214/aos/1013203451.

[14] Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22. doi:10.18637/jss.v033.i01.

[15] Gatto, L., Vanderaa, C., & Breckels, L. M. (2023). A foundation for reproducible single-cell and bulk proteomics data analysis in Bioconductor. *Nature Communications*, 14, 5698. doi:10.1038/s41467-023-41556-2.

[16] Gentleman, R., Carey, V. J., Huber, W., Irizarry, R. A., & Dudoit, S. (Eds.). (2005). *Bioinformatics and Computational Biology Solutions Using R and Bioconductor*. Springer, New York. doi:10.1007/0-387-29362-0.

[17] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer, New York, NY.

[18] Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67. doi:10.1080/00401706.1970.10488634.

[19] Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer, New York, NY. doi:10.1007/978-1-4614-6849-3.

[20] Kuhn, M., & Silge, J. (2022). *Tidy Modeling with R*. O'Reilly Media, Sebastopol, CA.

[21] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.

[22] Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S. I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–67. doi:10.1038/s42256-019-0138-9.

[23] Platt, J. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 10(3), 61–74.

[24] Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., & Smyth, G. K. (2015). *limma* powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*, 43(7), e47. doi:10.1093/nar/gkv007.

[25] Smyth, G. K. (2004). Linear models and empirical Bayes methods for assessing differential expression in microarray experiments. *Statistical Applications in Genetics and Molecular Biology*, 3(1), Article 3. doi:10.2202/1544-6115.1027.

[26] Strobl, C., Boulesteix, A. L., Zeileis, A., & Hothorn, T. (2007). Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics*, 8, 25. doi:10.1186/1471-2105-8-25.

[27] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1), 267–288. doi:10.1111/j.2517-6161.1996.tb02080.x.

[28] Vapnik, V. N. (1998). *Statistical Learning Theory*. John Wiley & Sons, New York.

[29] Vickers, A. J., & Elkin, E. B. (2006). Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*, 26(6), 565–574. doi:10.1177/0272989X06295361.

[30] Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2), 301–320. doi:10.1111/j.1467-9868.2005.00503.x.